

Story 2

Can the FED Control Inflation and Maintain Full Employment

PROF. PETER
KOWALCHUK

DATA-608

MEHREEN ALI
GILLANI



Introduction

The Federal Reserve operates under a dual mandate from Congress: **maintain price stability** (control inflation) and **promote maximum employment** (maintain low unemployment). These objectives often appear contradictory—policies that cool inflation (raising interest rates) can slow economic growth and increase unemployment, while policies that boost employment (lowering rates) can fuel inflation.

This assignment analyzes **25 years of economic data (1999-2024)** to answer a critical question: "**Has the Fed been able to fulfill the mandate given to it by Congress?**"

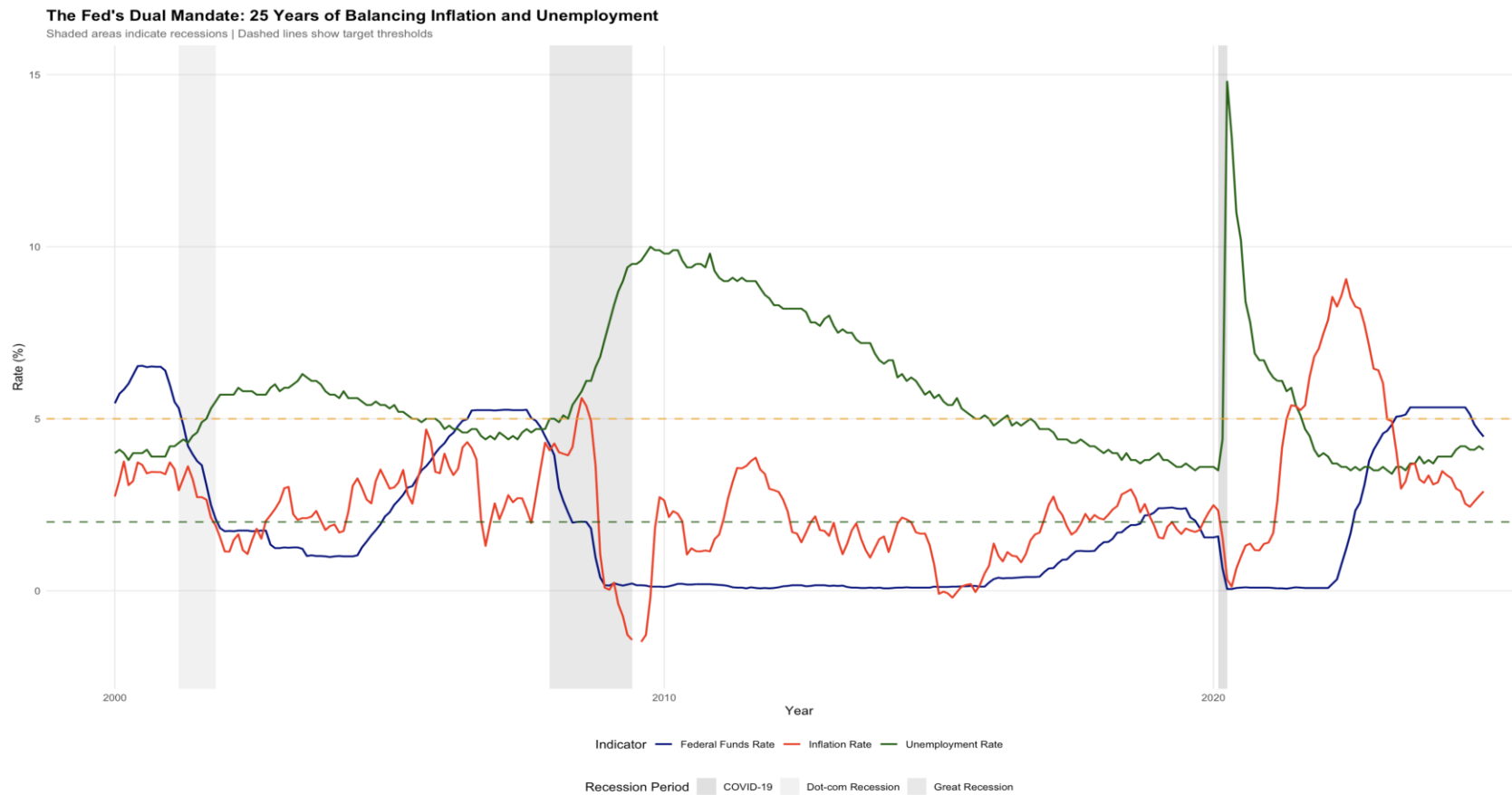
Data Sources

Data series	Source	API	Purpose
Consumer Price Index (CPI)	Bureau of Labor Statistics	BLS API	Measures inflation
Unemployment Rate	Bureau of Labor Statistics	BLS API	Measures Unemployment health
Federal Fund Rate	Federal Reserve (FRED)	FRED API	Measures Fed policy actions

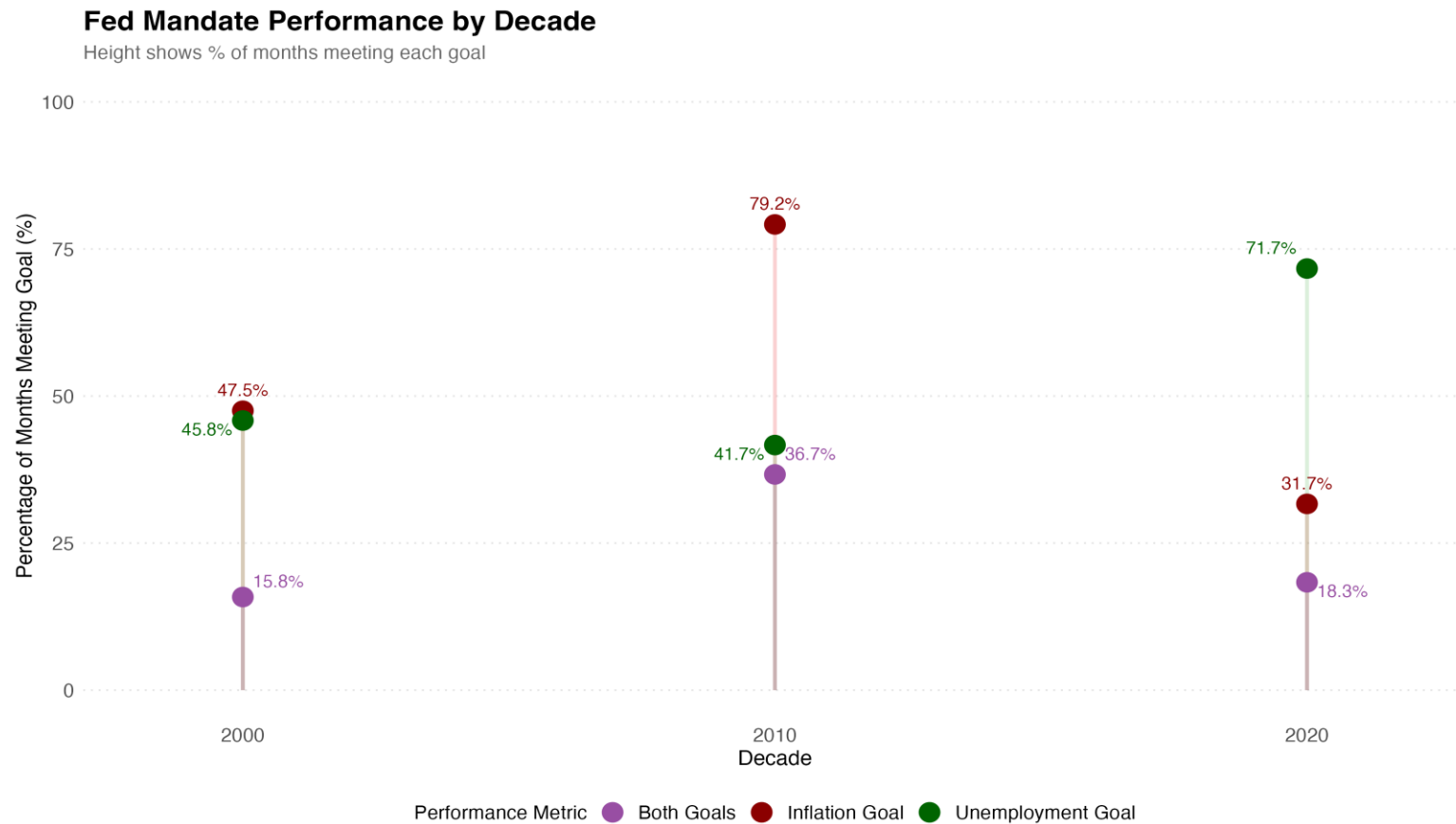
Methodology

- **Data Collection:** Accessed via APIs using Python, retrieving 312 monthly observations
- **Data Integration:** Combined all three series into a single dataset
- **Analysis:** Calculated year-over-year inflation, identified periods where goals were met
- **Visualization:** Created multi-panel plots in R showing:
 - Historical trends of all three indicators
 - Decade-by-decade mandate performance
 - The inflation-unemployment trade-off (Phillips Curve)
 - Causal chain showing lagged effects of rate hikes

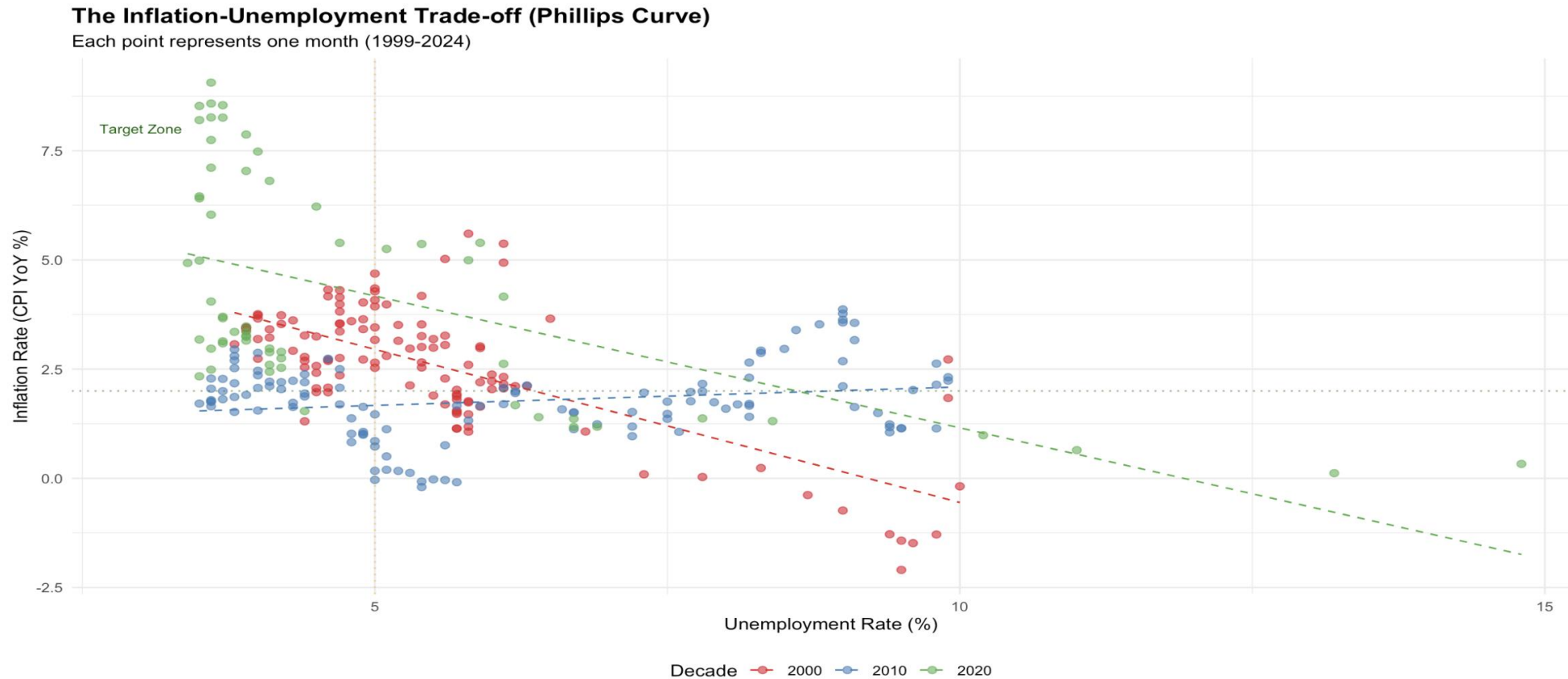
The Impossible Balance? Tracking the Fed's Twin Goals Since 1999



A Tale of Three Decades: Fed Mandate Performance Since 2000

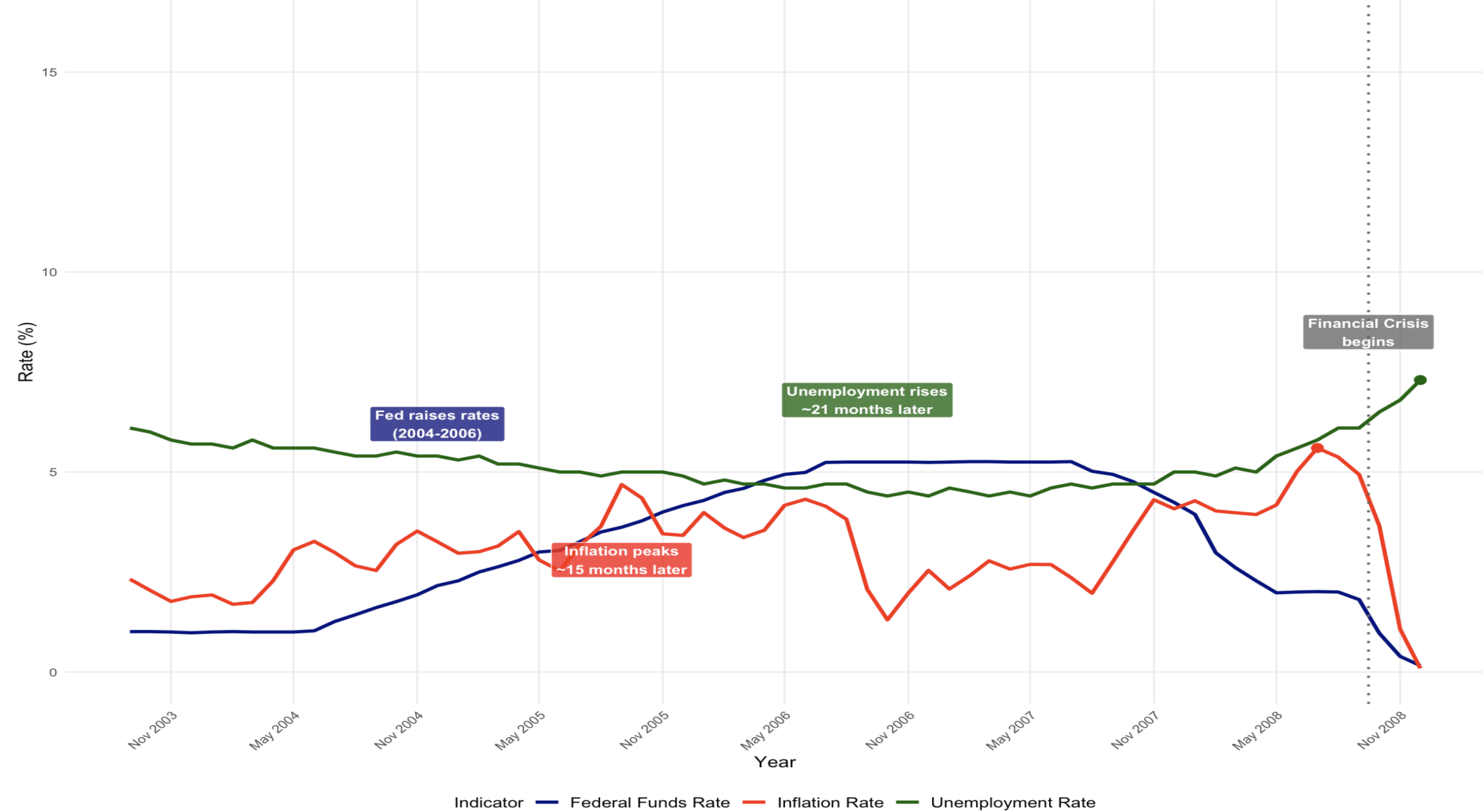


When One Goal Conflicts with Another: The Phillips Curve in Action



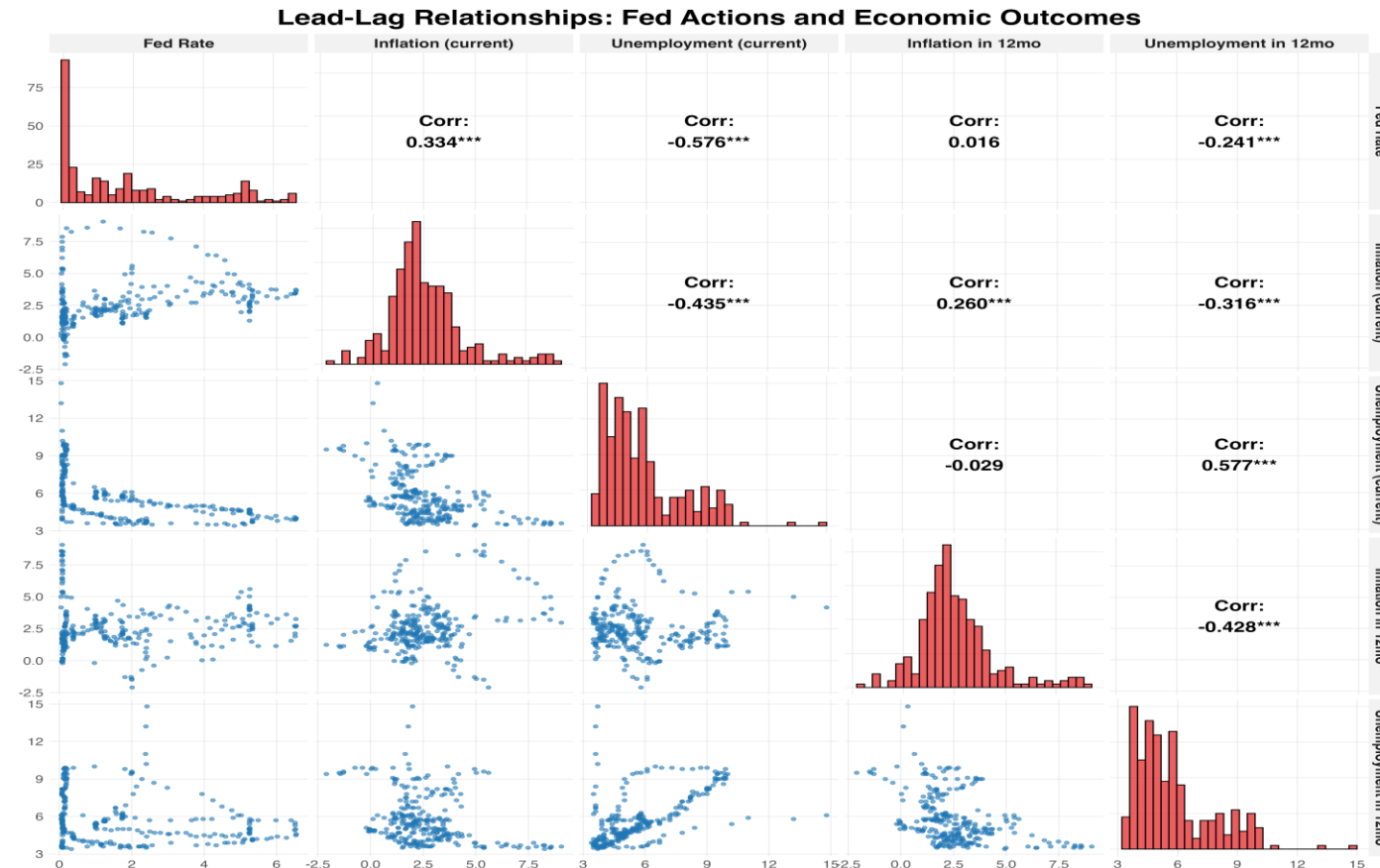
The Fed's Causal Chain: Rate Hikes → Lower Inflation → Higher Unemployment

2004-2008 cycle shows typical time lags | Inflation peaks before unemployment rises



Gray arrows show causal sequence | Dotted line marks Financial Crisis

The Ripple Effect: How Fed Policy Propagates Through the Economy



Conclusion

The Fed has been a capable crisis manager but an inconsistent mandate fulfiller. The dual mandate remains an aspirational target—achievable only about a quarter of the time—rather than a consistently attainable goal.

Recommendations for Policymakers

- Communicate the lag effect more clearly to the public
- Set realistic expectations that both goals cannot always be met
- Consider a flexible timeline for achieving both mandates rather than simultaneous success

"Monetary policy is a balancing act—and the Fed walks a tightrope between inflation and unemployment every single day."